

ECOSHIELD

Intelligence at the Edge of Every Disaster

A Resilient, AI-Powered Environmental Intelligence Network

Complete Project Document — Architecture, Hardware, AI/ML, Deployment and Feasibility

Smart India Hackathon 2026

Problem Statement 26178 — Qualcomm Inc.

Category: Hardware | Theme: Disaster Management

Team Name: CODE SMITHS

Team Members: Kasula Kiran, K.S.V.Dheeraj Kumar, B.Ram Charan Reddy, M.Divya Tejswi,
Ch.Jishnu Chowdary, K.Sai Sahitya Kannam

September 2026

Table of Contents

Document Status & Decision Legend	7
1. Executive Summary	7
1.1 What Makes EcoShield Different	7
2. Problem Statement and Background	8
2.1 Official Problem Statement	8
2.2 Hazards Addressed	8
2.3 Mandatory Requirements Mapped to Our Response	8
3. Existing Ecosystem and the Genuine Gap	9
3.1 Case Study: Kerala 2018 Floods and CAP Dissemination	9
4. Solution Overview and Design Principles	9
4.1 Vision	9
4.2 Design Principles (Locked)	9
5. System Architecture	10
5.1 Three-Tier Structure	10
5.2 End-to-End Data Flow	10
5.3 System Blueprint	10
6. Hardware Design (Student-Buildable Prototype)	11
6.1 Universal Core (built once, shared by every node)	11
6.2 Sensor Pod — Hydro / Flood	11
6.3 Sensor Pod — Fire / Air	11
6.4 Terrain / Landslide Pod (Advanced Extension)	12

6.5 Edge Gateway	12
6.6 Cost Summary and Build Recommendation	12
6.7 Build Order (2–3 Weekends)	12
7. Modular Sensor Pod System	12
7.1 Concept	12
7.2 Automatic Pod Identification	12
7.3 Safe Field-Replacement Procedure	13
7.4 Three Types of Replacement	13
7.5 Maintenance Logistics	13
8. Sensor, Node, Gateway, Network and System Health	13
9. Communication and Offline Resilience	14
9.1 Communication Path	14
9.2 Offline Behaviour by Condition	14
9.3 Payload Design	15
10. AI/ML Architecture	15
10.1 AI-0 — Node-Level Intelligence (offline-capable, low power)	15
10.2 AI-1 — Qualcomm/Edge Gateway Intelligence (primary environmental intelligence layer)	15
10.3 AI-2 — Cloud/Regional Intelligence (not on the emergency path)	16
10.4 Evidence Fusion Formula	16
10.5 Qualcomm / Edge Workload Placement	16
10.6 Detection vs. Forecasting — an Honest Distinction	16
11. Risk, Confidence and Alert Logic	16
11.1 Risk and Confidence as Separate Dimensions	16

11.2 Quorum-Gated Alerting	17
12. Camera and Computer Vision Pipeline	17
12.1 Pipeline Stages	17
12.2 Sustainability: E-Waste-Sourced Cameras	17
13. Hazard-Specific Intelligence	18
13.1 Flood Node Positioning	18
14. Security and Trust	18
14.1 Additional Security Measures	19
15. Alerting and Last-Mile Communication	19
15.1 Voice-Based Alerting	19
15.2 Human Confirm Alerts	19
15.3 Authority Acknowledgement and Escalation	19
15.4 CAP Bridge to SACHET / C-DOT	20
15.5 Recommended-Action Engine	20
16. GIS Dashboard, Impact Intelligence and Digital Twin	20
16.1 Digital Twin (Advanced Layer)	20
17. Deployment Strategy	20
17.1 Risk-Driven Node Placement	20
17.2 Gateway Zones and Redundancy	21
17.3 Representative Deployment Patterns	21
17.4 Infrastructure-Piggyback Deployment	21
17.5 Economic Framing: Cost per km ² of Coverage	21

18. Software Stack and Data Architecture	21
18.1 Software Stack	22
18.2 Event Data Model	22
19. Datasets and Model Lifecycle (MLOps)	22
19.1 Data Sources	22
19.2 Training, Validation and Deployment Pipeline	23
20. Feasibility, Risks and Mitigation	23
20.1 Feasibility	23
20.2 Challenges and Risks	23
20.3 Mitigation Strategies	23
20.4 Graceful Degradation Table	24
20.5 Additional India-Scale Limitations	24
21. Impact and Benefits	24
21.1 Impact on Target Audience	25
21.2 Impact Metrics (Design Targets — Pending Field Validation)	25
21.3 Benefits	25
22. Prototype Scope and Demonstration Plan	25
22.1 What Is Physically Built	25
22.2 What Is Simulated or Integrated Rather Than Fully Built	26
22.3 Live Demonstration Sequence	26
23. Roadmap	26
24. Innovation Summary	27
25. Anticipated Jury Questions and Prepared Answers	27

26. References 28

26.1 Government and Institutional Sources 28

26.2 Satellite and Scientific Data Sources 28

26.3 Hardware and Component References 28

26.4 Qualcomm and AI Platform References (Production Target) 29

26.5 ML and Networking Standards 29

Appendix A — Full Feature Checklist 30

Appendix B — Bill of Materials (Prototype) 32

Appendix C — Sample Event Payload 32

Appendix D — Glossary of Terms 33

Document Status & Decision Legend

Label	Meaning
LOCKED	Agreed architecture, feature, or design principle finalized by the team.
BENCHMARK-PENDING	Approach is fixed; exact model/component variant is selected after measured testing.
ADVANCED	Part of the target architecture; not a dependency of the core working prototype.
ROADMAP	Production-scale capability for post-hackathon deployment; not claimed as working today.
VERIFY BEFORE CLAIM	Value, cost, availability, or performance figure must be re-checked before stating as fact.

1. Executive Summary

India faces recurring, high-impact environmental hazards: urban and river floods across Assam, Bihar, Kerala and Maharashtra; forest fires across Uttarakhand, Himachal Pradesh and central India; hazardous air pollution in major cities; landslides in hill states; and industrial gas/chemical incidents. Government agencies such as NDMA, IMD and ISRO already run environmental monitoring and early-warning systems, but these are largely centralized and coarse-grained. They lack dense, hyperlocal, real-time, offline-capable sensing at the exact point of risk.

EcoShield is a distributed, AI-powered Environmental Intelligence Network built around a Universal Node Platform with field-replaceable, hazard-specific sensor pods. Each node performs on-device (edge) AI to detect anomalies locally. Nodes connect over LoRaWAN to an edge gateway, which performs advanced multi-sensor fusion, camera-based vision AI, and risk assessment — continuing to operate even when internet connectivity fails. Only critical alerts, summaries and risk assessments are transmitted onward to cloud infrastructure and, ultimately, to regional control centres and national dissemination systems such as NDMA's SACHET and C-DOT's CAP-based alerting.

Core Design Pipeline: Sense → Verify → Fuse → Predict → Decide → Alert → Act → Learn

The system is explicitly designed to complement — not replace — India's existing monitoring and dissemination ecosystem (IMD, CWC, FSI, CPCB, ISRO, SACHET/C-DOT). EcoShield's contribution is the missing layer: continuous, distributed, evidence-fused, offline-resilient local sensing and decision-making, feeding validated intelligence upward into systems that already reach the public at national scale.

1.1 What Makes EcoShield Different

- A single Universal Node hardware core that is reconfigured for any hazard by swapping a sensor pod — not five different products.
- Automatic pod identification and AI-profile loading — no firmware reflash needed when a pod is changed.
- Risk (severity) and Confidence are computed and reported as two separate values, not collapsed into one number.
- Quorum-Gated Alerting: a public CRITICAL alert requires corroboration from at least two independent evidence sources, which also functions as the system's core anti-spoofing defence.
- Trust-weighted neighbour recalibration, grounded in published wireless-sensor-network research, so a drifting sensor is corrected rather than simply ignored.
- A full offline-first emergency path — local detection, local voice/siren alerting, and store-and-forward buffering — that keeps working with zero internet connectivity.
- A five-layer security spine (signed identity, encrypted transport, physical-plausibility checks, quorum-gated fusion, tamper/audit logging) defending against cloned nodes, replay attacks and spoofed readings.

2. Problem Statement and Background

2.1 Official Problem Statement

Problem Statement ID: 26178 | Sponsor: Qualcomm Inc. | Category: Hardware | Theme: Disaster Management.

Official title: “A resilient, AI-powered environmental monitoring network that provides early detection, localized intelligence, and actionable alerts for floods, forest fires, pollution events, and other environmental hazards common in India, enabling authorities and communities to shift from reactive disaster response to proactive risk prevention.”

The description calls for a distributed Environmental Intelligence Network deployable across cities, rivers, forests, industrial zones and vulnerable communities, using on-device AI, multi-hazard early warning, regional risk mapping, community/authority notification, a cloud-edge hybrid architecture, and scalable, cost-effective deployment.

2.2 Hazards Addressed

- Urban flooding and river/flash floods — Assam, Bihar, Kerala, Maharashtra and other flood-prone states.
- Forest and vegetation fires — Uttarakhand, Himachal Pradesh, and central Indian forest belts.
- Hazardous air pollution — major urban centres.
- Landslides and terrain instability — hill and slope regions.
- Industrial gas leaks and chemical incidents — industrial zones and plant perimeters.
- Water quality degradation and extreme heat/weather — as advanced deployment extensions.

2.3 Mandatory Requirements Mapped to Our Response

PS Requirement	EcoShield Response
Distributed smart sensor nodes	Universal Node core + modular hazard-specific sensor pods (Hydro, Fire/Air, Terrain, Water Quality, Weather)
Solar-powered, low-maintenance	Solar + LiFePO4 battery, INA219 power telemetry, field-replaceable pods instead of full-node disposal
On-device AI, no continuous cloud dependency	AI-0 node intelligence + AI-1 edge gateway; the full emergency path works fully offline
Multi-hazard early warning	Shared event schema and hazard-specific models for flood, fire, pollution, landslide and water quality
Regional risk mapping	GIS dashboard with hotspots, trends, and population/infrastructure impact overlays
Community and authority notification	Dashboard, SMS, multilingual voice, IVR, local siren; severity and confidence shown separately
Cloud and edge hybrid architecture	Three-tier AI: Node (AI-0) → Edge Gateway (AI-1) → Cloud (AI-2)
Scalable, cost-effective deployment	Universal core + swappable pods, gateway zones, infrastructure-piggyback siting
IoT protocol support	LoRaWAN (IN865) node-to-gateway; 4G/5G/Wi-Fi/Ethernet gateway-to-cloud

3. Existing Ecosystem and the Genuine Gap

EcoShield is designed to complement, not replace, India's existing monitoring and alerting infrastructure. The genuine gap it targets is the combination of distributed local sensing, on/near-device intelligence, sensor-health awareness, multi-sensor and neighbour evidence fusion, offline resilience, and actionable geo-targeted risk intelligence — not the absence of any single government system.

System	What It Contributes	EcoShield's Relationship
CWC / IMD flood warning	Hydrometeorological observations, forecasts, institutional flood warning	Complements with localized field sensing, rate-of-rise detection and edge decisions
FSI / satellite fire alerts (MODIS/VIIRS)	Large-area fire hotspot detection, roughly 3-hour latency between satellite passes	Complements with ground-level smoke/PM/temperature evidence and faster local verification
CPCB air-quality monitoring	Continuous, reference-grade ambient air-quality data at fixed stations	Complements with distributed, low-cost, denser local observations
ISRO / remote sensing	Regional and spatial environmental context	Used as optional external evidence, not the sole real-time input
NDMA SACHET / C-DOT CAP	Geo-targeted, multi-channel dissemination of authorized disaster alerts (SMS, cell broadcast, apps, TV, radio)	EcoShield feeds validated, CAP-compatible alerts into this system rather than rebuilding national dissemination

3.1 Case Study: Kerala 2018 Floods and CAP Dissemination

Kerala's official Orange Book of Disaster Management records that C-DOT's Common Alerting Protocol-compliant platform, integrated with telecom providers for geo-targeted SMS, was used to send up to 11.4 lakh precautionary messages during the 2018 floods. This is strong evidence that the dissemination problem can be solved operationally at scale — but it is evidence of dissemination capability, not of field-level hazard detection. EcoShield is designed to feed richer, earlier, evidence-backed alert events into exactly this kind of dissemination infrastructure.

Note: Quantitative figures such as the 11.4 lakh message count are historical (2018) and should be labelled with their time period; C-DOT's platform has since evolved.

4. Solution Overview and Design Principles

4.1 Vision

EcoShield is an evidence-fused, edge-first, resilient environmental intelligence platform. It senses multiple hazards, verifies sensor reliability, fuses local and external evidence, detects and assesses risk at the edge, and delivers resilient, multi-channel alerts even when cloud connectivity is unavailable.

4.2 Design Principles (Locked)

- Local intelligence first — critical decisions never depend on continuous cloud connectivity.
- Evidence over single-sensor triggers — one abnormal reading never automatically becomes a public disaster alert.
- Risk and confidence are treated as separate dimensions, not one blended score.
- Sensor health is itself a form of environmental intelligence.

- One common node platform serves every deployment environment through modular sensor pods and AI profiles.
- The network complements existing government and institutional systems rather than claiming to replace them.
- Advanced features are part of the target architecture, but all prototype claims are limited to what is actually built and measured.
- Human authorities remain responsible for final safety decisions and operational actions — the system is decision support, not autonomous control.

5. System Architecture

5.1 Three-Tier Structure

Layer	Primary Role	Example Hardware
Sensor Node	Acquire data, local filtering, health checks, local buffering, LoRaWAN transmission	ESP32-S3-class MCU + modular sensor pod
Edge Gateway	Multi-node fusion, advanced AI, camera vision, risk engine, local alerts, backhaul	Raspberry Pi (prototype) / Qualcomm Dragonwing IQ-8275 (production target)
Cloud / Regional Control	Historical analytics, GIS, model training and lifecycle, fleet management	Backend API + time-series database + GIS dashboard

5.2 End-to-End Data Flow

Node → LoRaWAN → Edge Gateway → 4G/5G/Wi-Fi/Ethernet → Cloud/Control Centre → CAP/SACHET → Public and Authority Channels.

- Normal operation: nodes send compact telemetry summaries; the gateway fuses and forwards them; the cloud updates GIS and analytics.
- Internet unavailable: the gateway continues local inference, risk assessment and local alerting; events are buffered.
- Gateway unavailable: the node keeps measuring, applies its own local safety logic, stores events, and retries/synchronizes once the gateway returns.
- Critical extreme event: a deterministic safety override can bypass normal evidence-gathering delay when a reading is far beyond any plausible danger threshold.

5.3 System Blueprint

EXTERNAL INTELLIGENCE (Weather / Satellite / Existing Systems) feeds into the Node layer alongside direct sensing. Universal Nodes (sensor pods, sensor health, local baseline, trend/anomaly detection, safety rules, buffering) send data over LoRaWAN to the Edge Gateway (multi-node fusion, advanced ML, camera/thermal AI, risk + confidence engine, local alerts, offline operation). The gateway forwards over 4G/5G/Wi-Fi to the Cloud/Regional Control Centre (GIS, history, training, forecasting, digital twin, fleet and model registry), which in turn feeds the CAP/SACHET alert ecosystem (SMS, voice, cell broadcast, app, radio, TV, web).

6. Hardware Design (Student-Buildable Prototype)

The hardware design deliberately favours widely available, inexpensive, well-documented components so that a student team can source, wire and validate a full working prototype without specialised equipment. Every component listed below is commercially available in India at the time of writing; exact prices should be re-verified before purchase.

6.1 Universal Core (built once, shared by every node)

Component	Part	Approx. Cost (₹)	Role
MCU	ESP32-S3 (WROOM-1, N16R8)	700 – 900	Dual-core 240MHz, 512KB SRAM, Wi-Fi/BLE; runs filtering and TinyML
LoRaWAN radio	Seeed Wio-E5 Dev Kit	3,500 – 3,800	Pre-certified IN865 module, AT-command firmware
Power monitor	TI INA219 breakout	150 – 250	I2C battery/solar current-voltage-power telemetry
Battery	LiFePO4 cell, ~6Ah (e.g. IFR26650)	500 – 700	Safer than Li-ion outdoors, better heat tolerance
Solar panel	6V/9V, 5–10W panel	400 – 700	Trickle-charges the battery in typical Indian sun
Charge controller	TP5000 or equivalent LiFePO4-compatible module	150 – 300	Handles charging without custom analog design
Enclosure	IP65 ABS junction box (150x110x70mm)	250 – 400	Weatherproof housing, field-drillable cable glands

Universal Core subtotal: approximately ₹6,000 – 7,500

6.2 Sensor Pod — Hydro / Flood

- DFRobot A02YYUW — waterproof ultrasonic water-level sensor, IP67, 3–450cm range, UART (₹1,600–1,800)
- DFRobot SEN0575 — tipping-bucket rainfall sensor (₹1,200–2,000)
- DHT22 / SHT31 — temperature and humidity sensor (₹250–500)

Hydro Pod subtotal: approximately ₹3,000 – 4,300 | Full Hydro Node (Core + Pod): ₹9,000 – 11,800

6.3 Sensor Pod — Fire / Air

- Sensirion SPS30 — PM1/PM2.5/PM4/PM10 particulate sensor, laser scattering, I2C/UART (₹3,800–4,200)
- MQ-2 — smoke / combustible gas sensor, analog output (₹150–300)
- DHT22 / SHT31 — temperature and humidity sensor (₹250–500)
- MLX90640 32x24 thermal array — optional, for visual fire confirmation (₹3,000–4,500)

Fire/Air Pod subtotal: approximately ₹4,200 – 5,000 (excluding thermal) | Full Fire/Air Node (Core + Pod): ₹10,200 – 12,500

6.4 Terrain / Landslide Pod (Advanced Extension)

- LSM6DSOX / BMI270 IMU — tilt and vibration sensing (prototype); ADXL355 for production-grade precision
- Capacitive or RS-485 soil-moisture probe
- Shared temperature/humidity sensor from the same DHT22/SHT31 family

6.5 Edge Gateway

- Raspberry Pi 4 or 5 (4GB sufficient) — ■5,000–8,000 — runs Python/MQTT/fusion logic
- USB LoRa receiver (a second Wio-E5 in USB-serial mode, or an RFM95-based USB dongle) — ■2,000–3,000
- Raspberry Pi Camera Module 3 — optional, ■2,500–3,000, for the camera-AI demonstration

Gateway total: approximately ■7,000 – 14,000 depending on camera inclusion

Note: Production target for the edge gateway is the Qualcomm Dragonwing IQ-8275 (20–40 dense INT8 TOPS, heterogeneous CPU/GPU/NPU/MCU, MIPI CSI, industrial temperature rating). The Raspberry Pi is presented explicitly as a prototype/fallback platform, not as equivalent hardware.

6.6 Cost Summary and Build Recommendation

- One Universal Core + both pods (proves live modularity): approximately ■13,200 – 16,800
- Full demoable prototype including gateway: approximately ■20,000 – 31,000

Recommendation: build one Universal Core with both sensor pods and physically swap the pod during the live demonstration, rather than building two entirely separate nodes. This is both cheaper and a stronger proof of the modularity claim than two static units.

6.7 Build Order (2–3 Weekends)

- Weekend 1 — establish ESP32-S3 to Wio-E5 UART communication; send and receive a test LoRa packet on the gateway. Confirms the communications backbone before any sensor work.
- Weekend 2 — wire and validate Hydro pod sensors individually; confirm sane readings; integrate into firmware with a CUSUM-based persistence check.
- Weekend 3 — wire and validate Fire/Air pod sensors; integrate the node-level fire-score logic; build the pod-identification mechanism.

7. Modular Sensor Pod System

7.1 Concept

The node is designed as a rugged, field-hardware version of plug-and-play modularity: a Universal Core that never changes, combined with hazard-specific Sensor Pods that can be added, removed or replaced. This is deliberately “LEGO-like” in principle but industrial-grade in execution — keyed, locking, weatherproof, vibration-resistant connectors, not consumer-toy connectors.

7.2 Automatic Pod Identification

- Each sensor pod carries its own small identity/calibration memory chip (a low-cost I2C EEPROM), storing pod type, model, measurement range, unit, calibration coefficients and driver/firmware compatibility.
- The moment a pod is connected, the node reads this chip over the shared I2C bus and instantly knows what is attached — no manual configuration.
- The node matches the identity against an internal Pack Registry (a lookup table) mapping pod type to sensor drivers, AI model file, and alert thresholds.
- It then automatically initializes the correct drivers, loads the matching pre-trained AI model from flash storage, applies calibration coefficients, and reports READY status to the dashboard.
- No firmware reflash is required — the same physical node becomes a Flood node, Fire node or Air-Quality node purely based on which pod is attached.

7.3 Safe Field-Replacement Procedure

Hot-swapping is explicitly not claimed as a safety feature, since outdoor sensors are exposed to moisture, static discharge, and electrical transients that make live swapping unsafe. Instead, the platform uses a controlled Maintenance Mode sequence:

Maintenance Mode → isolate sensor interface → remove old/failed pod → insert replacement pod → identify pod metadata → self-test → calibration/health validation → load driver + AI profile → return to service.

7.4 Three Types of Replacement

- Sensor replacement — same pod type, e.g. a broken PM sensor pod swapped for a new one of the same type.
- Sensor expansion — adding an additional pod to an existing node, e.g. a flood node gaining a water-quality pod.
- Mission reconfiguration — changing what the node monitors entirely, e.g. a forest fire node relocated and reconfigured as an industrial gas node.

7.5 Maintenance Logistics

Because the expensive components (MCU, radio, battery system, enclosure) remain with the Universal Core, a maintenance team only needs to carry spare pods rather than entire spare nodes — significantly simplifying field logistics at scale.

8. Sensor, Node, Gateway, Network and System Health

Health monitoring is treated as a first-class intelligence dimension, not an afterthought. The platform must be able to distinguish a real environmental event from a failing sensor, degraded node, communication problem or power problem — and it must never let equipment failure masquerade as an environmental disaster.

Health Layer	Monitored Signals
Sensor health	Range plausibility, missing data, stuck readings, impossible jumps, temporal consistency, calibration state, self-test
Node health	Battery/charge state, radio signal quality, last-seen time, storage utilization, firmware version, device temperature
Gateway health	Backhaul status, LoRaWAN service status, connected-node count, queue/backlog, CPU/memory/load, power/UPS state
Network health	Packet-loss rate, latency, node freshness, gateway reachability, reconnect attempts

Health Layer	Monitored Signals
System health state	HEALTHY / DEGRADED / MAINTENANCE / OFFLINE — kept explicitly separate from environmental risk

Unhealthy sensors are down-weighted or excluded from evidence fusion according to configured rules, rather than being silently trusted. Maintenance alerts are generated for calibration overdue, battery degradation, repeated sensor faults, communication deterioration, and tamper events. The fleet dashboard displays health and environmental risk as separate views so an operator can immediately distinguish “high hazard” from “unreliable node.”

9. Communication and Offline Resilience

9.1 Communication Path

Node → LoRaWAN (IN865 regional plan) → Edge Gateway → 4G/5G/Wi-Fi/Ethernet → Cloud/Control Centre.

- LoRaWAN is the primary field telemetry link because node payloads are small and low power consumption is essential.
- The gateway carries its own cellular SIM, giving it an independent voice/SMS/IVR path that functions even when broadband backhaul is down.
- Node data does not require continuous raw streaming to the cloud — normal operation transmits compact summaries; events and critical evidence receive higher transmission priority.

9.2 Offline Behaviour by Condition

Condition	System Behaviour
Cloud available	Node sends telemetry/event summaries → gateway → cloud → dashboard and external alert services
Internet unavailable	Gateway continues local inference, risk assessment and local alerting; events are buffered
Gateway unavailable	Node keeps measuring, applies available local safety logic, stores events, retries/syncs when gateway returns
Critical extreme event	Deterministic safety override bypasses normal persistence waiting when a reading is far beyond plausible danger threshold

9.3 Payload Design

Data Type	Content	Frequency Strategy
Normal telemetry	Node ID, timestamp, sensor values, battery, RSSI, health	Low rate / compressed
Watch event	Feature vector + risk state + top evidence	Higher rate
Critical event	Risk, confidence, evidence, health, sequence, location, local action status	Immediate
Camera evidence	Inference metadata by default; image/clip only on high-confidence events	Event-driven, via 4G/5G/Wi-Fi — never over LoRa
Offline data	Buffered records with sequence numbers	Uploaded after reconnection

Note: Mesh self-healing (a node automatically re-associating with the nearest live gateway if its primary gateway fails) is listed as a ROADMAP feature only. It is not claimed as a working prototype capability unless the reassociation logic is actually implemented and tested.

10. AI/ML Architecture

The AI architecture is deliberately distributed across three tiers with distinct responsibilities — not three copies of the same model. Model selection throughout is benchmark-driven: a simpler model is preferred whenever it matches or beats a more complex one on measured precision, recall, false-alarm rate, latency, memory and energy.

10.1 AI-0 — Node-Level Intelligence (offline-capable, low power)

Signal processing chain: raw sensor reading → range/plausibility check → Median/Hampel filter (spike removal) → EWMA/Kalman smoothing → rate-of-change and acceleration features → CUSUM sustained-shift detection → quantized TinyML logistic classifier → deterministic safety override for extreme values.

CUSUM formula: $C_t = \max(0, C_{t-1} + (x_t - \text{baseline} - k))$; an anomaly is flagged when C_t exceeds a configured threshold h . This detects a sustained shift in sensor readings while remaining robust to single-sample noise.

10.2 AI-1 — Qualcomm/Edge Gateway Intelligence (primary environmental intelligence layer)

- Isolation Forest — unsupervised anomaly scoring, chosen because real disaster-event labels are naturally sparse and imbalanced.
- XGBoost — separate supervised hazard classifiers per hazard type (Flood-XGB, Fire-XGB, Pollution-XGB, Landslide-XGB); efficient enough to run on CPU, no NPU required.
- GRU — short-horizon escalation forecasting (e.g. predicted water level 10 minutes ahead); benchmarked against LSTM and simple lag-feature baselines rather than assumed superior.
- Explicit spatial-graph features — neighbour_mean_risk, upstream/downstream gradient, agreement ratio — used instead of a full graph neural network in the prototype.
- RTMDet → MobileNetV3-Small — a compact vision detector identifies a smoke/flame region of interest, and a lightweight verifier classifies the cropped region; both models are supported on Qualcomm AI Hub for the IQ-8275 target.

10.3 AI-2 — Cloud/Regional Intelligence (not on the emergency path)

- Model training, calibration and validation on larger historical datasets.
- GAT/GraphSAGE and temporal-GNN research for advanced hazard-propagation modelling — explicitly an advanced research direction, not a prototype prerequisite.
- Digital twin and what-if simulation.
- Fleet analytics, drift monitoring, and a versioned model registry with rollback.

10.4 Evidence Fusion Formula

$E = \Sigma(w \cdot e) / \Sigma(w)$, where $w = \text{base_reliability} \times \text{data_quality} \times \text{freshness} \times \text{relevance}$.

Every evidence source — a sensor reading, a neighbouring node's corroboration, weather context, or camera evidence — is dynamically weighted rather than treated equally. A stale, low-quality or historically unreliable source contributes proportionally less to the final risk assessment.

10.5 Qualcomm / Edge Workload Placement

The architecture deliberately uses heterogeneous compute rather than assuming every model must run on an AI accelerator: XGBoost and Isolation Forest run efficiently on general-purpose CPU; GRU, RTMDet and MobileNetV3-Small use the NPU; image preprocessing uses the ISP/GPU where available. On the Raspberry Pi prototype, all models run on CPU; NPU placement is the documented production plan for the Qualcomm Dragonwing IQ-8275.

10.6 Detection vs. Forecasting — an Honest Distinction

A single low-cost node with only its own sensors can realistically perform anomaly/threshold detection — recognizing that something unusual is happening right now at its location. Genuine multi-hour forecasting (e.g. “75% chance of flooding in 3 hours”) requires broader context: data from multiple nodes across a basin, historical patterns, and often external rainfall forecasts. EcoShield therefore places detection at the node (AI-0) and short-to-medium horizon forecasting at the gateway and cloud tiers (AI-1/AI-2), where that broader context is actually available.

11. Risk, Confidence and Alert Logic

11.1 Risk and Confidence as Separate Dimensions

Risk State	Meaning	Primary Response
NORMAL	No meaningful abnormality	Monitoring only
WATCH	Unusual trend or early evidence; insufficient corroboration for a public warning	Dashboard / internal monitoring
HIGH	Multiple evidence sources support an active/developing hazard	Dashboard + SMS/app/web notification
CRITICAL	Strong evidence and/or severe, rapidly escalating conditions	Multi-channel emergency alert + authority escalation + local warning

Confidence (LOW / MEDIUM / HIGH) is computed and displayed independently of severity. A “CRITICAL / MEDIUM confidence” state is an intentional, explainable possibility — a potentially severe reading that lacks full independent

corroboration is not the same as a “HIGH / HIGH confidence” event that is strongly supported by multiple sources. This separation is deliberately kept visible on the dashboard as a live two-axis (quadrant) view rather than collapsed into a single number.

11.2 Quorum-Gated Alerting

A CRITICAL public alert requires corroboration from at least two independent evidence sources (for example, two sensors on the same node, a neighbouring node, or camera confirmation), or an extreme, deterministic safety-override threshold. This single design rule serves two purposes simultaneously: it is the escalation logic that prevents false public alarms from a single noisy sensor, and it is the system's core defence against a single spoofed or compromised node fabricating a disaster.

12. Camera and Computer Vision Pipeline

The camera is treated strictly as a gateway-side visual module, not a LoRa sensor. Raw video is never transmitted over LoRaWAN — the link's bandwidth simply cannot carry it, and attempting to do so would defeat the purpose of a low-power sensor network.

12.1 Pipeline Stages

- Capture — RGB (Raspberry Pi Camera Module 3) and optionally thermal (FLIR Lepton 3.5 / MLX90640) frames at the gateway.
- Local transfer — camera connects directly to the gateway via MIPI CSI or USB.
- Pre-processing — resize/crop/normalize, frame-rate reduction, motion/event gating.
- Vision inference — RTMDet detects a smoke/flame region of interest; MobileNetV3-Small verifies the cropped region.
- Evidence extraction — detection class, confidence, timestamp, location, bounding box, model version.
- Multimodal fusion — visual evidence combined with PM, gas, temperature, humidity, neighbour and weather evidence.
- Upstream transmission — compact event metadata by default; a compressed snapshot or short clip is uploaded only for high-confidence events, via 4G/5G/Wi-Fi — never LoRaWAN.
- Offline mode — the gateway continues vision inference, fusion and local alerting even when backhaul fails; selected evidence is buffered for later synchronization.

12.2 Sustainability: E-Waste-Sourced Cameras

Where feasible, camera modules are sourced from discarded electronics rather than purchased new. This simultaneously reduces prototype hardware cost and gives the project a genuine circular-economy / sustainability angle that is uncommon among comparable systems.

13. Hazard-Specific Intelligence

Hazard	Core Features	Advanced Features
Flood / flash flood	Water level, rate of rise, acceleration, rainfall, local baseline, upstream/downstream nodes	CWC/IMD context, satellite rainfall, soil saturation, terrain/inundation context
Forest / vegetation fire	PM2.5/PM10, gas/smoke, temperature, humidity, temporal change, neighbour nodes	RGB/thermal vision, wind/dryness, satellite hotspot evidence, spatial spread
Air pollution	PM2.5/PM10, gas/VOC/CO, temperature/humidity, baseline, trend	Weather/wind, neighbour nodes, source-region inference
Landslide / terrain instability	Soil moisture, tilt/IMU, vibration, rainfall, persistence, spatial agreement	Terrain/slope data, remote sensing, historical event context
Water quality degradation	pH, turbidity, conductivity, temperature, persistence	Dissolved oxygen, source metadata, downstream propagation

13.1 Flood Node Positioning

Flood nodes follow the hydrological path rather than an arbitrary grid, using four positional classes:

- Upstream node — earliest detection of a rising water event.
- River/channel node — tracks the main flow.
- Downstream node — confirms progression toward vulnerable areas.
- Vulnerable-zone node — placed directly at the community or infrastructure at risk.

This upstream → channel → downstream → vulnerable-zone chain allows the system to measure the spatial progression of a flood wave and warn downstream nodes before the water arrives, rather than relying on a single absolute water-level reading.

14. Security and Trust

Security is treated as a full defence-in-depth spine, where each layer defends against a specific, named attack rather than a generic “we have security” claim.

Layer	Defence	Stops
Identity	Unique signed device ID (ATECC608-class secure element target), authenticated provisioning	Fake/cloned node impersonating a real one
Transport	Encrypted LoRaWAN session keys + sequence counters	Replayed old packets, man-in-the-middle injection
Physical plausibility (AI-0)	Range/rate-of-change checks at the node itself	Obviously impossible spoofed values (e.g. water level jumping 3m in one reading)
Quorum-gated fusion (AI-1)	A CRITICAL alert requires ≥ 2 independent evidence sources or an extreme safety-override threshold	A single hacked or spoofed node trying to fake a disaster

Layer	Defence	Stops
Audit + tamper	Every event traceable to node ID + model version + evidence trace; tamper-switch/IMU flags physical interference	After-the-fact forensics, physical tampering detection

14.1 Additional Security Measures

- Role-based access control for dashboard users.
- Audit logs for every alert and every operator action.
- Firmware integrity checks and a signed/controlled firmware update process.
- Safety-critical models are never self-modifying in production; updates follow train → validate → approve → deploy.

15. Alerting and Last-Mile Communication

Risk Level	Alert Channels
NORMAL	No emergency alert
WATCH	Dashboard / internal monitoring
HIGH	Dashboard + SMS + app/web notification
CRITICAL	Dashboard + SMS + voice + IVR (where deployed) + local siren/display + authority escalation

15.1 Voice-Based Alerting

- Pre-validated emergency message templates, with dynamic fields limited to safe values (location/zone, severity, action category).
- Multilingual support — prototype target languages: English, Hindi and Telugu.
- Templates are preloaded on the gateway so critical local voice alerts do not depend on cloud text-to-speech.

15.2 Human Confirm Alerts

Residents can confirm what they are observing via a simple IVR keypress or app action. This human corroboration feeds directly into the confidence score alongside sensor and camera evidence, closing the loop between the automated network and the people who actually live in the affected area.

15.3 Authority Acknowledgement and Escalation

Alert status progresses through generated → delivered → acknowledged → resolved. Unacknowledged CRITICAL events automatically escalate to the next configured authority level. Full audit logs preserve alert and acknowledgement history.

15.4 CAP Bridge to SACHET / C-DOT

The Alert Engine includes a CAP Bridge that converts an approved internal event into a Common Alerting Protocol-compliant message — hazard type, severity, certainty/confidence representation, location/geometry, effective time, expiry, instruction, and language fields. Internal rich evidence (raw sensor values, model scores, node health, fusion trace) is kept separate from the public alert payload, but every public alert references its originating event ID and model/rule version for full auditability. EcoShield positions itself explicitly as complementing SACHET/C-DOT dissemination infrastructure, not replacing or rebuilding it.

15.5 Recommended-Action Engine

The platform can surface rule-based, safety-reviewed suggested actions (e.g. monitor a downstream flood zone, verify a fire-affected area, inspect a bridge, issue an air-quality advisory). This layer is explicitly decision support — it does not autonomously issue unsafe operational commands.

16. GIS Dashboard, Impact Intelligence and Digital Twin

- Live regional GIS risk map with hazard hotspots and trends.
- Node status, health, and last-seen time.
- Population and critical-infrastructure impact overlays (hospitals, schools, bridges, roads, power/industrial assets).
- Evidence/explanation panel showing why a given alert was raised.
- Historical event timeline and alert acknowledgement/escalation status.

16.1 Digital Twin (Advanced Layer)

The digital twin is a what-if simulation layer (e.g. “simulate 200mm rainfall over this basin in 6 hours”) used for planner decision support and scenario testing. It is explicitly treated as an advanced, cloud-side capability rather than a headline feature or a prerequisite for the core working prototype, since equivalent functionality is commonly claimed by many competing systems.

17. Deployment Strategy

17.1 Risk-Driven Node Placement

Node density and placement are not based on a simplistic rule such as “one node every X metres.” Instead, placement follows a combined optimisation:

Risk Map + Hazard Map + Population + Critical Infrastructure + Terrain + Radio Coverage → Node Placement Optimisation.

- High-risk areas receive denser and/or redundant sensing.
- Low-risk areas use sparser coverage.
- Critical locations can have independent redundant sensors to improve corroboration.
- Placement aims to maximise useful evidence, not simply maximise node count.

17.2 Gateway Zones and Redundancy

Large deployments are divided into local gateway zones, each serving a cluster of nearby nodes. Critical zones can use overlapping gateway coverage so that loss of a single gateway does not eliminate local sensing. The resilience claim is explicitly edge decision-making plus local buffering plus retry/store-and-forward — a custom self-healing mesh is not claimed unless actually implemented and tested.

17.3 Representative Deployment Patterns

Environment	Node Pattern	Gateway Pattern
City	Dense multi-hazard nodes at drains, pollution hotspots, industrial boundaries	Rooftop / municipal / control-centre gateways
River basin	Upstream → channel → downstream → vulnerable-zone nodes	Elevated secure points with reliable backhaul
Forest	Distributed fire/air nodes along high-risk zones and boundaries	Elevated tower/ranger/secure gateway points
Hills	Slope nodes + lower-zone impact nodes	Stable elevated locations with good radio geometry
Industrial	Perimeter and process-risk nodes	Secure plant/control-room gateways, redundant coverage
Rural/vulnerable community	Hazard-specific nodes + local voice/siren capability	Solar/secure community or telecom-site gateway

17.4 Infrastructure-Piggyback Deployment

Wherever possible, sensor pods are retrofitted onto existing infrastructure — electricity poles, telecom towers, and CWC gauge stations — instead of constructing new greenfield sites. This sharply reduces installation cost, time, and land/permission overhead.

17.5 Economic Framing: Cost per km² of Coverage

Rather than presenting only a raw per-node bill of materials, deployment economics are framed as cost per km² of previously unmonitored coverage gained — turning the hardware cost table into an impact argument rather than a parts list.

18. Software Stack and Data Architecture

18.1 Software Stack

Layer	Technology	Purpose
Node firmware	ESP-IDF / Arduino C++, ESP-NN/ESP-DSP	Drivers, filtering, health, local inference, LoRa payloads, sleep/wake
Gateway services	Python, FastAPI, MQTT, OpenCV, ONNX Runtime / Qualcomm AI Runtime	Fusion, inference, camera pipeline, risk/alerts
Edge storage	SQLite / local PostgreSQL subset	Store-and-forward buffering and local history
Cloud API	FastAPI	Data and event APIs
Cloud database	PostgreSQL + TimescaleDB	Time-series sensor data + relational metadata
ML	scikit-learn, XGBoost, PyTorch	Training, benchmarking, temporal/spatial experiments
Model lifecycle	MLflow + object storage + approval registry	Versioning and controlled deployment
Frontend	Next.js/React + Leaflet/Mapbox	GIS risk map, node health, alert centre
Alerts	SMS/IVR provider + local voice/siren + CAP adapter	Multi-channel last-mile dissemination

18.2 Event Data Model

The event object is the core unit of downstream intelligence, retaining full evidence and health context rather than a single flag:

```
{ node_id, gateway_id, timestamp, location, hazard, risk_level, confidence, evidence[], trend, sensor_health, system_health, model_version, local_action_status, sequence_id }
```

19. Datasets and Model Lifecycle (MLOps)

19.1 Data Sources

Hazard	Primary Sources	Use
Flood	CWC hydrometeorological data, IMD gridded rainfall, NASA GPM (supplementary)	Historical features/labels and validation
Fire	NASA FIRMS (MODIS/VIIRS), FSI fire alerts, D-Fire dataset, IIITDMJ Smoke dataset	Event labels and vision model training/validation
Air quality	CPCB AQI/CAAQMS repository + meteorology	PM/gas baselines and spatio-temporal patterns
Landslide	NASA Global Landslide Catalog + rainfall/terrain data	Event labels and contextual features
Own network	Controlled bench data + field observations + operator feedback	Calibration, local baseline, fine-tuning, false-alarm analysis

Public and simulated datasets support model development but do not replace local calibration or prove field accuracy. Final claims must clearly distinguish real field measurements, public historical data, and simulated/controlled data.

19.2 Training, Validation and Deployment Pipeline

Data → QA → Label → Train → Validate → Calibrate → Register → Optimize → Edge Benchmark → Approval → Deploy → Monitor → Drift Detection → Retrain.

- Location-aware and time-aware evaluation splits are used to avoid data leakage from near-identical events at the same location.
- Reported metrics include precision, recall, F1, false-positive rate, false-negative rate, detection latency, lead time, inference time, model size and power.
- A deterministic extreme-value safety override is always maintained, even when ML confidence is uncertain.
- Safety-critical models are never self-modifying in production; every update goes through validation and approval before deployment.

20. Feasibility, Risks and Mitigation

20.1 Feasibility

- Built entirely on proven, commercially available components — not speculative hardware.
- Realistic prototype cost of approximately ■9,000–12,500 per fully instrumented node, and ■20,000–31,000 for a complete two-pod demonstration build including gateway.
- Infrastructure-piggyback deployment sharply reduces installation cost and time versus greenfield sites.
- A phased rollout — prototype → pilot → district → state — with each phase gated by measured validation rather than assumption.
- Benchmark-driven model selection throughout; no algorithm is chosen because it “sounds advanced.”

20.2 Challenges and Risks

- LoRa range collapses in dense forest and hilly terrain due to foliage and elevation attenuation.
- The gateway is a potential single point of failure for its node cluster.
- Low-cost gas/PM sensors drift over time and require periodic recalibration at scale.
- A safety-critical alert network is an attractive target for spoofed or malicious node injection.
- Field maintenance and labour logistics across remote, climatically diverse terrain.
- Live integration with CAP/SACHET depends on authorized government access.
- Class imbalance in training data, since real disaster events are naturally rare.

20.3 Mitigation Strategies

- Ridge-line gateway placement, directional antennas, and redundant gateway coverage in critical zones.
- Store-and-forward buffering plus a local voice/siren failsafe independent of gateway/cloud availability; the gateway's own SIM provides a backup voice/SMS/IVR path even without broadband.

- Trust-weighted neighbour recalibration turns sensor drift into a self-correcting network property instead of a maintenance emergency.
- The five-layer security spine directly defends against spoofing and cloned-node attacks; no single compromised node can trigger a public CRITICAL alert.
- A two-tier labour model: local caretakers (aligned with Panchayat/Jal Jeevan Mission networks) handle basic upkeep, while regional technicians perform pod-level field replacement, guided by remote health telemetry so visits are planned rather than reactive.
- A CAP-compatible payload is built and demonstrated now, with live SACHET/C-DOT integration explicitly positioned as an authorized deployment step rather than a prototype dependency.
- Weighted losses, event-based evaluation, and careful negative sampling address class imbalance honestly rather than through synthetic overfitting.

20.4 Graceful Degradation Table

Failure	System Response
One sensor fails	Marked degraded; down-weighted or excluded in fusion; other sensors continue
One node fails	Neighbour nodes/backup coverage continue; event loss logged
Gateway loses internet	Local AI, local alerts and buffering continue unaffected
Gateway fails entirely	Nodes continue local safety logic, buffer, and retry/sync when a gateway returns
Camera fails	Sensor-only path remains available; confidence adjusted accordingly
Cloud unavailable	Zero effect on immediate local emergency decisions
Power low	Adaptive sensing reduces duty cycle; maintenance health state escalates
False sensor anomaly	Sensor-health engine prevents it becoming an automatic public alert

20.5 Additional India-Scale Limitations

- Spectrum/regulatory compliance — LoRa in India operates under WPC rules; NB-IoT requires telecom operator spectrum agreements.
- Extreme terrain and climate diversity — coastal salinity, Himalayan sub-zero winters, and desert heat each demand different hardware variants.
- Multi-agency governance — disaster response is constitutionally a state subject, requiring routing to the correct SDMA/district authority.
- Data ownership, privacy and liability for camera-based monitoring near private property.
- Manufacturing and supply-chain resilience at volumes beyond hobbyist-grade components.
- Cybersecurity of a safety-critical network at national scale.

21. Impact and Benefits

21.1 Impact on Target Audience

Rural and urban communities in flood-prone (Assam, Bihar, Kerala, Maharashtra), fire-prone (Uttarakhand, Himachal Pradesh, central India) and industrially exposed regions gain earlier, localized, offline-resilient warnings — shifting disaster response from reactive to proactive. Human Confirm Alerts and multilingual offline voice ensure the system reaches people without smartphones or reliable data connectivity. Government agencies (NDMA, IMD, ISRO, state SDMA) gain a dense, hyperlocal evidence layer that strengthens their existing forecasting and dissemination infrastructure rather than duplicating it.

21.2 Impact Metrics (Design Targets — Pending Field Validation)

- Minutes-level local detection versus approximately 3 hours for satellite-based fire detection (NASA FIRMS MODIS/VIIRS); satellites can miss short-lived fires entirely between overpasses.
- Reduced false-alarm rate via quorum-gated, multi-sensor and neighbour-corroborated alerting versus single-sensor triggers.
- Zero dependency on cloud connectivity for the local emergency alerting path.
- Deployment cost reduction via infrastructure-piggyback siting versus greenfield installation.
- Scalable from a single node (~₹9,000–12,500) to district/state coverage without redesigning hardware.
- Eight distinct failure modes explicitly handled with a defined system response, not just the happy path.

21.3 Benefits

Social

Faster, localized, spoof-resistant warnings save lives in remote and low-connectivity areas. Multilingual offline voice and IVR-based Human Confirm Alerts include populations that smartphone-app-based systems structurally leave behind.

Economic

Substantially lower cost than expanding reference-grade infrastructure (a single CPCB monitoring station costs ₹50 lakh or more). Infrastructure-piggyback siting and cost-per-km²-of-coverage framing make network expansion economically defensible. Earlier, better-verified action reduces downstream disaster response and relief costs.

Environmental

E-waste-sourced camera modules reduce both electronic waste and hardware cost simultaneously. Better forest-fire, pollution and water-quality monitoring directly supports conservation and public-health policy. Solar-powered, low-maintenance nodes minimize the network's own environmental footprint.

22. Prototype Scope and Demonstration Plan

22.1 What Is Physically Built

- One Universal Core with a field-replaceable Hydro pod and a field-replaceable Fire/Air pod.
- LoRaWAN link using Wio-E5 in IN865 configuration.
- One edge gateway (Raspberry Pi, with camera).
- Sensor-health and false-sensor demonstration.

- Flood escalation demonstration.
- Fire evidence-fusion demonstration.
- Internet-disconnect / offline demonstration with buffered-event synchronization on reconnect.
- Critical voice/local-warning demonstration.
- GIS dashboard with a representative multi-node deployment.

22.2 What Is Simulated or Integrated Rather Than Fully Built

- District/state-scale deployment and full gateway redundancy.
- Live satellite and national weather feed integration.
- Full production C-DOT CAP/SACHET integration — a standards-compatible CAP payload/adaptor is demonstrated instead.
- A full state-scale digital twin — a small synthetic flood/fire propagation scenario is demonstrated instead.
- Large-scale retraining pipelines — versioned training/validation on public and controlled data is demonstrated instead.
- Production-grade industrial radar level sensors and precision gas/soil probes.

22.3 Live Demonstration Sequence

- 1. Normal state — dashboard remains NORMAL.
- 2. Controlled flood stimulus — escalates through WATCH to HIGH/CRITICAL with visible trend evidence.
- 3. Controlled fire stimulus — temperature/PM/gas plus camera evidence fuse into a single event.
- 4. False sensor case — an isolated abnormal reading is classified as a sensor anomaly, not a public alert.
- 5. Disconnect the internet — the edge gateway continues risk calculation and local alerting.
- 6. Reconnect — buffered events synchronize to the cloud.
- 7. Live pod swap — the Hydro pod is removed and the Fire/Air pod is inserted; the node auto-identifies and reconfigures with zero reflash.
- 8. Critical event — voice/local warning plus the alert acknowledgement workflow.

23. Roadmap

Phase	Scope	Gate for Progression
Prototype	2–3 field nodes, one edge gateway, core hazards, offline and alert demo	Repeatable readings + working local decisions + measured comms/power
Pilot	Small region, real sensor observations, representative gateway coverage	False-alarm rate, calibration quality, uptime validated
District	More nodes, redundant gateways at critical zones, external-data integration	Coverage, reliability, maintenance logistics validated
State	Multi-zone deployment, regional GIS, authorized CAP/SACHET integration	Operational governance, data/security approvals, sustained performance
Multi-state / National	Fleet management, placement optimization, broader dissemination	Long-term reliability, validated models, governance and economics

24. Innovation Summary

Ranked by defensibility — how difficult each feature would be for a competing team to replicate quickly, and how directly each answers a question a technical jury is likely to ask.

- Quorum-Gated Alerting — simultaneously the escalation logic and the core anti-spoofing defence.
- Trust-Weighted Neighbour Recalibration — grounded in published wireless-sensor-network research (Takruri et al., spatio-temporal drift-aware calibration), not a guess.
- Risk separated from Confidence as two independently computed and displayed values.
- Five-layer Security Spine — identity, transport, plausibility, quorum fusion, audit/tamper.
- Human Confirm Alerts — resident corroboration feeding directly into the confidence score.
- Universal Node + Field-Replaceable Pods with automatic identification and zero-reflash AI-profile loading.
- E-waste-sourced camera modules — a genuine circular-economy differentiator.
- Gateway's own SIM — an independent voice/SMS/IVR path even without broadband.
- Infrastructure-piggyback deployment and cost-per-km² framing — turning the BOM into an economic argument.
- Camera AI processed entirely at the edge, with raw video never traversing LoRaWAN.

Digital twin simulation and cloud-side GenAI assistance are retained in the architecture as advanced capabilities, but are deliberately not headlined, since equivalent functionality is commonly claimed by many competing systems and does not differentiate EcoShield on its own.

25. Anticipated Jury Questions and Prepared Answers

- Why does the node need AI if the gateway has more compute? — The node's AI-0 layer exists to keep the emergency detection and local-alert path functional with zero connectivity to the gateway or cloud; it performs filtering, plausibility checks and a lightweight anomaly signal so the network degrades gracefully rather than going blind.
- Why not send all data directly to the cloud? — LoRaWAN payloads are extremely small by design to keep power consumption low; continuous raw streaming is neither technically possible over this link nor necessary, since most value comes from summarized, evidence-fused decisions rather than raw time-series.
- Why can't LoRa carry camera video? — LoRa payloads are measured in bytes, not kilobytes; a single video frame is orders of magnitude too large. This is precisely why the camera connects directly to the gateway rather than routing through a node's LoRa radio.
- How do you distinguish a faulty sensor from a real hazard? — Every reading passes through sensor-health scoring (plausibility, temporal consistency, calibration state) and is checked against neighbouring nodes; an isolated abnormal reading without corroboration is flagged as a sensor anomaly rather than promoted to a public alert.
- How do you prevent a false alarm from one sensor? — Quorum-Gated Alerting requires at least two independent evidence sources, or a deterministic extreme-value override, before any CRITICAL public alert is issued.
- How does the system work without internet? — The edge gateway performs the full risk/confidence calculation and issues local voice/siren alerts entirely offline; events are buffered locally and synchronized once connectivity returns.
- What happens when a sensor pod is replaced? — The new pod's identity chip is read automatically, the correct driver and AI profile are loaded, and the node completes a self-test and calibration check before returning to service — no firmware reflash is required.

- What data trains your models? — A combination of public institutional datasets (CWC, IMD, CPCB, NASA FIRMS, NASA Global Landslide Catalog) for baseline features and labels, and the team's own controlled bench and field data for calibration and fine-tuning; these sources are always clearly distinguished in reporting.
- How do you evaluate false negatives and false positives? — Using location-aware and time-aware train/test splits to avoid leakage, and reporting precision, recall, F1, false-positive rate, false-negative rate, detection latency and lead time for every candidate model before selection.
- What is simulated versus physically demonstrated? — The physical prototype demonstrates two real field nodes, one gateway, live pod-swap reconfiguration, offline/reconnect behaviour, and evidence-fused alerting; district/state-scale deployment, live SACHET integration, and a full digital twin are explicitly simulated or represented as standards-compatible interfaces rather than claimed as fully built.
- Is a large language model making the safety decision? — No. Generative AI is restricted to natural-language dashboard queries, human-readable explanations of why an alert was raised, and report summarization. It is never the primary hazard detector and does not independently decide whether a critical event exists.

26. References

26.1 Government and Institutional Sources

- SIH 2026 PS 26178 — Qualcomm Inc. — <https://sih2026.vuce.in/ps/SIH26178>
- NDMA SACHET Portal — <https://sachet.ndma.gov.in/>
- C-DOT CAP Integrated Alert System — <https://www.cdor.in/cdotweb/assets/docs/products/dms/cap.pdf>
- Kerala SDMA, Orange Book of Disaster Management — <https://sdma.kerala.gov.in/>
- CWC flood management / hydrometeorological data — <https://cwc.gov.in/>
- CPCB AQI / CAAQMS data repository — https://airquality.cpcb.gov.in/AQI_India/
- FSI forest fire alert system — <https://fsiforestfire.gov.in/>
- ISRO Disaster Management Support — <https://www.isro.gov.in/DisasterManagementSupport.html>

26.2 Satellite and Scientific Data Sources

- NASA FIRMS (MODIS/VIIRS active fire data) — <https://earthdata.nasa.gov/data/tools/firms>
- NASA Global Landslide Catalog — <https://earth.gsfc.nasa.gov/hydro/data/landslide-hazard-assessment>
- NASA GPM precipitation products (supplementary flood data)

26.3 Hardware and Component References

- Espressif ESP32-S3 — <https://www.espressif.com/en/node/4993>
- Seeed Wio-E5 (IN865 LoRaWAN) — https://wiki.seeedstudio.com/LoRa_E5_Dev_Board/
- Sensirion SPS30 — <https://sensirion.com/sps30>
- DFRobot A02YYUW — <https://wiki.dfrobot.com/sen0311>
- DFRobot SEN0575 rainfall sensor — <https://wiki.dfrobot.com/sen0575/>
- Analog Devices ADXL355 — <https://www.analog.com/en/products/adxl355.html>
- Raspberry Pi Camera Module 3 — <https://www.raspberrypi.com/products/camera-module-3/>

- Teledyne FLIR Lepton — <https://oem.flir.com/en-in/products/lepton/>

26.4 Qualcomm and AI Platform References (Production Target)

- Qualcomm Dragonwing IQ-8275 — <https://www.qualcomm.com/internet-of-things/products/iq8-series/iq-8275>
- Qualcomm IQ-8275 Evaluation Kit — <https://www.qualcomm.com/developer/hardware/qualcomm-iq-8275-evaluation-kit-evk>
- Qualcomm AI Hub — RTMDet — <https://aihub.qualcomm.com/iot/models/rtdet>
- Qualcomm AI Hub — MobileNetV3-Small — https://aihub.qualcomm.com/iot/models/mobilenet_v3_small

26.5 ML and Networking Standards

- XGBoost documentation — <https://xgboost.readthedocs.io/en/stable/>
- LoRa Alliance — LoRaWAN Specification v1.0.3 — <https://resources.lora-alliance.org/document/lorawan-specification-v1-0-3>
- Takruri et al. — spatio-temporal drift-aware WSN calibration (basis for trust-weighted neighbour recalibration) — exact title/DOI to be verified before final submission

Note: All prices, availability, and performance figures throughout this document are engineering planning values captured during design and must be re-verified from primary sources before being stated as final facts in a submission or to a jury.

Appendix A — Full Feature Checklist

Feature	Status
Universal Core (common MCU, power, storage, radio, identity, firmware)	Core — Locked
Modular sensor pods (Hydro/Fire/Air/Terrain/Water-Quality/Weather)	Core — Locked
Automatic pod identification + calibration metadata	Core — Locked (headline)
Automatic AI-profile loading on pod swap (no reflash)	Core — Locked (headline)
Safe field-replacement procedure (maintenance mode)	Core — Locked
Hot-swapping	Explicitly NOT claimed
Solar + LiFePO4 power, INA219 telemetry	Core — Locked
E-waste-sourced camera modules	Unique differentiator
5-layer health monitoring (sensor/node/gateway/network/system)	Core — Locked
LoRaWAN IN865 node-to-gateway	Core — Locked
4G/5G/Wi-Fi/Ethernet gateway-to-cloud	Core — Locked
Gateway's own SIM for independent voice/SMS/IVR	Core — Locked
Store-and-forward offline buffering	Core — Locked
Mesh self-healing / gateway failover	Roadmap / P2 only
Three-tier AI: Node → Edge → Cloud	Core — Locked
Sensor-health-aware evidence fusion	Core — Locked
Neighbour-node corroboration	Core — Locked
Quorum-Gated Alerting	Unique differentiator
Trust-Weighted Neighbour Recalibration	Unique differentiator (research-grounded)
Risk separated from Confidence	Unique differentiator
Adaptive sampling by risk state	Core — Locked
Camera/thermal vision at edge, metadata only over LoRa	Core — Locked
Benchmark-driven model selection	Core — Locked

Feature	Status
Dashboard, SMS, voice, IVR, local siren	Core — Locked
Alert acknowledgement + escalation workflow	Core — Locked
Human Confirm Alerts	Unique differentiator
CAP-compatible payload/adaptor for SACHET/C-DOT	Core — Locked
Recommended-action engine (decision support only)	Core — Locked
GIS risk map, population/infra overlay	Core — Locked
Digital twin / what-if simulation	Advanced — mention once
Security spine (identity/transport/plausibility/quorum/audit)	Core — Locked
Continual learning with train/validate/approve/deploy gate	Core — Locked
Controlled GenAI (explanations, NL queries)	Advanced — not primary detector
Risk-driven node placement	Core — Locked
Gateway zones + critical-zone redundancy	Core — Locked
Infrastructure-piggyback deployment	Unique differentiator
Cost-per-km ² coverage framing	Unique differentiator

Appendix B — Bill of Materials (Prototype)

Item	Approx. Cost (■)
ESP32-S3 (WROOM-1, N16R8)	700 – 900
Seeed Wio-E5 Dev Kit	3,500 – 3,800
TI INA219 power monitor	150 – 250
LiFePO4 cell, ~6Ah	500 – 700
Solar panel, 5–10W	400 – 700
TP5000 charge controller	150 – 300
IP65 enclosure	250 – 400
DFRobot A02YYUW ultrasonic water-level sensor	1,600 – 1,800
DFRobot SEN0575 rain gauge	1,200 – 2,000
Sensirion SPS30 PM sensor	3,800 – 4,200
MQ-2 gas/smoke sensor	150 – 300
DHT22 / SHT31 temperature-humidity sensor (x2)	500 – 1,000
Raspberry Pi 4/5 (gateway)	5,000 – 8,000
USB LoRa receiver	2,000 – 3,000
Raspberry Pi Camera Module 3 (optional)	2,500 – 3,000
Total — full demoable prototype (Core + both pods + gateway)	20,000 – 31,000

Appendix C — Sample Event Payload

```
{
  "node_id": "HYDRO-014",
  "gateway_id": "GW-04",
  "timestamp": "2026-09-27T11:42:10+05:30",
  "location": { "lat": 26.14, "lon": 91.73 },
  "hazard": "flood",
  "risk_level": "HIGH",
  "confidence": "MEDIUM",
  "evidence": ["water_level_rate", "rainfall_trend", "neighbor_agreement"],
  "sensor_health": "HEALTHY",
  "system_health": "HEALTHY",
  "model_version": "flood-xgb-v1.3",
  "local_action_status": "siren_active",
```

"sequence_id": 88231

}

Appendix D — Glossary of Terms

Term	Meaning
AI-0 / AI-1 / AI-2	The three intelligence tiers: node, edge gateway, and cloud
CAP	Common Alerting Protocol — a standard alert message format used by SACHET/C-DOT
CUSUM	Cumulative Sum — a statistical method for detecting a sustained shift in a data stream
Evidence Fusion	Combining multiple independent evidence sources into a single weighted risk assessment
Quorum-Gated Alerting	Requiring corroboration from ≥ 2 independent sources before a CRITICAL public alert
Sensor Pod	A field-replaceable module carrying the sensors for one hazard type
Universal Core	The common hardware base (MCU, power, radio, storage) shared across all node types
Quorum	The minimum number of independent evidence sources required to confirm an event